

15.6 f

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15.6 Using the NLS panel data on $N = 716$ young women, we consider only years 1987 and 1988. We are interested in the relationship between $\ln(WAGE)$ and experience, its square, and indicator variables for living in the south and union membership. Some estimation results are in Table 15.10.

TABLE 15.10		Estimation Results for Exercise 15.6			
	(1)	(2)	(3)	(4)	(5)
	OLS 1987	OLS 1988	FE	FE Robust	RE
<i>C</i>	0.9348 (0.2010)	0.8993 (0.2407)	1.5468 (0.2522)	1.5468 (0.2688)	1.1497 (0.1597)
<i>EXPER</i>	0.1270 (0.0295)	0.1265 (0.0323)	0.0575 (0.0330)	0.0575 (0.0328)	0.0986 (0.0220)
<i>EXPER</i> ²	-0.0033 (0.0011)	-0.0031 (0.0011)	-0.0012 (0.0011)	-0.0012 (0.0011)	-0.0023 (0.0007)
<i>SOUTH</i>	-0.2128 (0.0338)	-0.2384 (0.0344)	-0.3261 (0.1258)	-0.3261 (0.2495)	-0.2326 (0.0317)
<i>UNION</i>	0.1445 (0.0382)	0.1102 (0.0387)	0.0822 (0.0312)	0.0822 (0.0367)	0.1027 (0.0245)
<i>N</i>	716	716	1432	1432	1432

(standard errors in parentheses)

f. Column (5) contains the random effects estimates. Which coefficients, apart from the intercepts, show the most difference from the fixed effects estimates? Use the Hausman test statistic (15.36) to test whether there are significant differences between the random effects estimates and the fixed effects estimates in column (3) (Why that one?). Based on the test results, is random effects estimation in this model appropriate?

$$t = \frac{b_{FE,k} - b_{RE,k}}{[\hat{v}ar(b_{FE,k}) - \hat{v}ar(b_{RE,k})]^{1/2}} = \frac{b_{FE,k} - b_{RE,k}}{[se(b_{FE,k})^2 - se(b_{RE,k})^2]^{1/2}} \quad (15.36)$$

Column (5) reports the random effects estimates. Compared with the fixed effects estimates in column (3), the coefficients that show the largest differences, apart from the intercept, are *SOUTH* and *EXPER*.

The coefficient on *SOUTH* changes from -0.3261 under fixed effects to -0.2326 under random effects. The difference is

$$-0.3261 - (-0.2326) = -0.0935. \quad (1)$$

The coefficient on *EXPER* changes from 0.0575 under fixed effects to 0.0986 under random effects. The difference is

$$0.0575 - 0.0986 = -0.0411. \quad (2)$$

Thus, in absolute coefficient terms, *SOUTH* shows the largest difference.

The Hausman test is used to test whether the individual-specific effect u_i is correlated with the regressors. The null hypothesis is

$$H_0 : Cov(u_i, X_{it}) = 0. \quad (3)$$

Under the null hypothesis, both the fixed effects and random effects estimators are consistent, but random effects is more efficient. If the null hypothesis is rejected, the random effects estimator is inconsistent, and fixed effects should be preferred.

Using equation (15.36),

$$t = \frac{b_{FE,k} - b_{RE,k}}{\sqrt{se(b_{FE,k})^2 - se(b_{RE,k})^2}}, \quad (4)$$

and comparing column (5) with column (3), we obtain the following Hausman t-statistics:

$$\begin{aligned} t_{EXPER} &\approx -1.67, \\ t_{EXPER^2} &\approx 1.30, \\ t_{SOUTH} &\approx -0.77, \\ t_{UNION} &\approx -1.06. \end{aligned} \quad (5)$$

The largest absolute t-statistic is for *EXPER*, but even this is smaller than the usual 5% critical value of 1.96 in absolute value. Therefore, we do not reject the null hypothesis that the individual effects are uncorrelated with the regressors.

We compare the random effects estimates with column (3), not column (4), because the simple Hausman test formula uses the usual estimated variances of the fixed effects and random effects estimators. Column (4) reports cluster-robust standard errors, and using those would require a robust version of the Hausman test rather than the simple formula in equation (15.36).

Based on the Hausman test results, random effects estimation appears to be appropriate for this model. There is evidence of individual heterogeneity from the earlier F-test, but there is not strong evidence that this heterogeneity is correlated with the explanatory variables. Therefore, random effects can be used, and it is more efficient than fixed effects if the null hypothesis is true.